Decode the market performance “gene map” of game products

MKT6019 Group 5

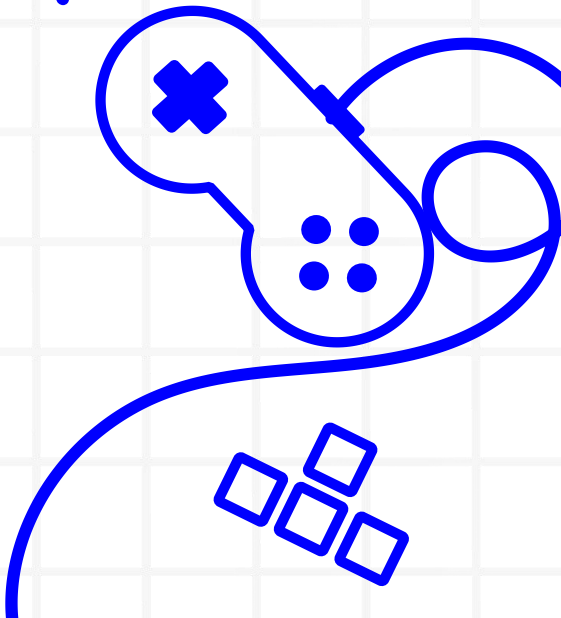


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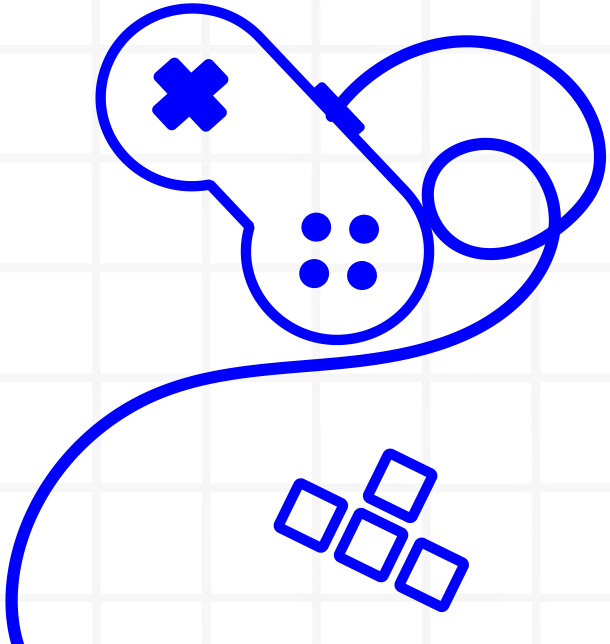
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01

Introduction

Launching the "Genetic Revolution" in the Game Market

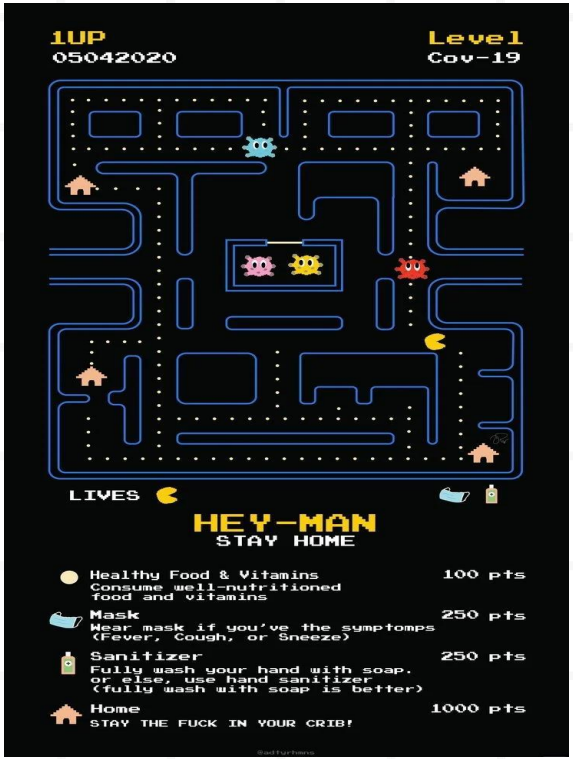
Research Background

The Gaming Industry's "Genetic Awakening"

The current game market faces unprecedented challenges: **severe product homogenization, rising user acquisition costs, and increasingly fierce competition.**

Against this backdrop, **understanding the fundamental mechanisms of game success and extracting patterns from data** has become the key to victory. Our Gene Decoder was born to meet this very challenge.

What is video games?



Pixel game Pac-Man



console game mario



Arcade games

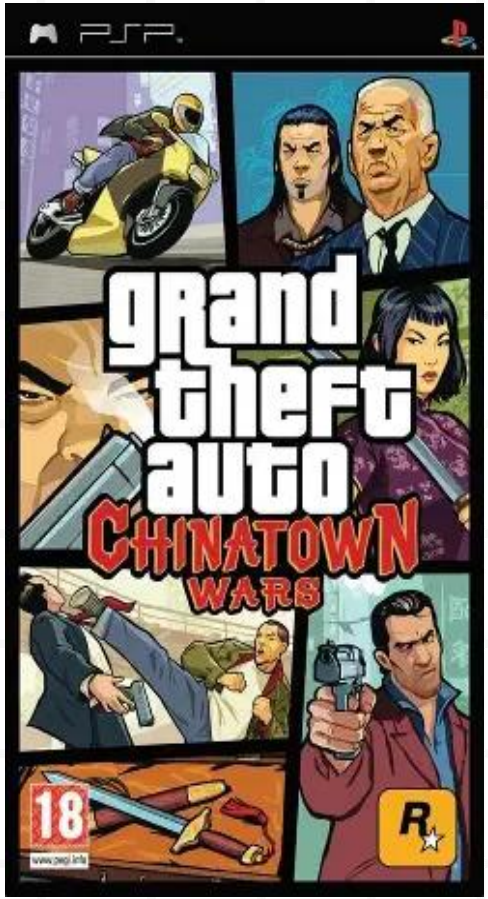


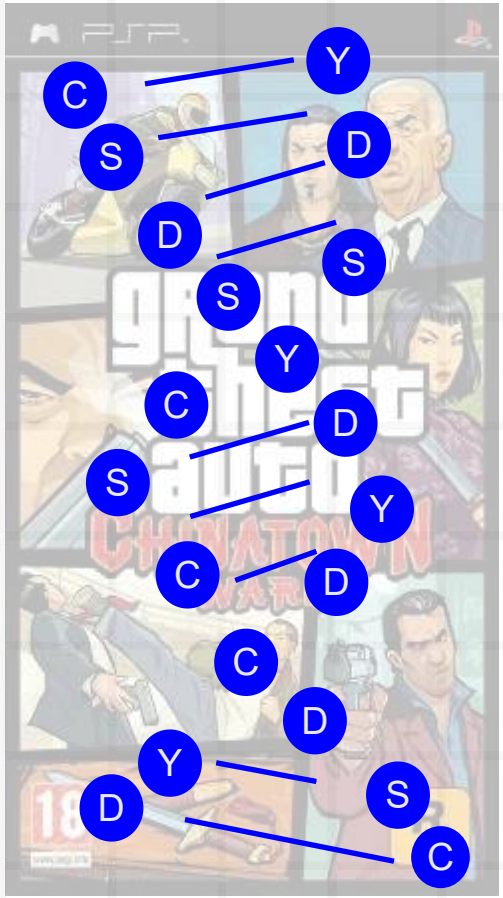
Chinese first 3A

What factors have made these games so popular and successful?

Imagine if we could "sequence the genes" of a game like a living organism?

relying on feelings? or looking at genes





Every game is a entity with a unique combination of genes.

The commercial success of a game is the result of the expression of its inherent "market genes".

Our job is to build a "gene decoder" to crack these genetic DNA codes.

Research Questions

Search for the "four major genetic codes" of game success

Signal Genes	How does a game's "trailer effect" before release influence its market performance? Do quality signals have a critical threshold?
Synergy Genes	How do brand combinations create a "chemical reaction"-like value-added effect? Which pairings generate super-linear returns?
Diffusion Genes	How do games trigger "viral transmission" within player communities? How do propagation patterns differ across regions?
Competition Genes	How can a game find its survival space in the fierce "battlefield"? How can we identify blue ocean opportunities?

Research Questions

What signals, coordination and competition mechanisms jointly drive the sales performance of games?

Market performance = f(signal gene × diffusion gene × Synergistic gene ×

02

Literature Review

Following the “Genetic laws"

Literature Review

Signaling Theory

- Reduces information asymmetry; publishers send "signals" (brand, price, ratings) (Spence, 1973; Kirmani & Rao, 2000).
- Media ratings & publisher reputation boost sales (Hsu et al., 2013; Zhang & Li, 2020).

Diffusion of Innovations

- Explains how games spread over time; innovators vs. imitators (Rogers, 2003; Bass, 1969).
- Regional adoption patterns highlight need for localization (Chen & Xie, 2008).

Brand Synergy & Network Effects

- Alliances create "1+1>2" value; publisher–developer partnerships enhance success (Aaker & Keller, 1990; Kumar, 2014).
- Network analysis identifies central actors in collaboration (Park & Kim, 2023).

Literature Review

Market Competition & Blue Ocean Strategy

Evolution theory- "Survival of the fittest"

- Compete via positioning or create new markets (Porter, 1985; Kim & Mauborgne, 2005).
- Player behavior & segmentation reveal underserved niches (Wu et al., 2020; Zhou & Wang, 2022).

Hofstede's Cultural Dimensions

Class of "game genes"

- Culture shapes adoption & preferences; six quantifiable dimensions (Hofstede, 1980).

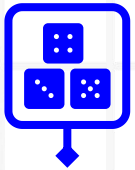
Power distance	Individualism	Masculinity
Uncertainty avoidance	long term orientation	indulgence

Theoretical Framework

Three-Layer Drive Model







	Driving Factors	Theoretical Basis	Analysis Method
Micro Level	User Ratings, Media Ratings, Number of Raters	Signaling Theory	Multiple Regression, Random Forest
Meso Level	Publisher Reputation, Developer Reputation, Brand Synergy	Brand Alliance Theory	Synergy Effect Score, Interaction Term Regression
Macro Level	Game Genre, Regional Diffusion, Market Competition	Diffusion of Innovations Theory, GNN	K-means Clustering, Bass Model, GNN

03

Data Description

Establishing a "Game Gene Sample Bank"

Data Collection

"Gene Bank" of the Gaming Industry

Dataset Source global regional data

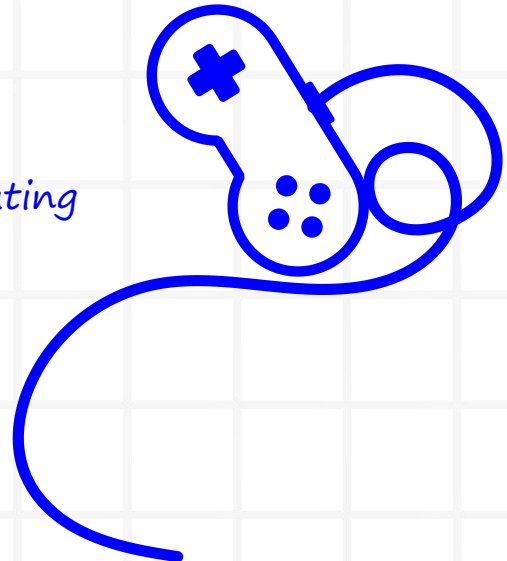
- Kaggle — "PlayStation Sales and Metadata (PS3, PS4, PS5)"
- Combines:
  - **VGChartz** → Global sales data
  - **RAWG API** → Game metadata

Data Overview abundant

- **4,963** PlayStation game products
- Platforms: **PS3, PS4, PS5**
- Multi-dimensional data: Sales performance, Review & rating information, Publisher & release details

Time Coverage new

- **PS3 / PS4**: Updated to **end of 2020**
- **PS5**: Data as of **October 2025** (limited by recency)



Data Processing

① Converted to Numeric

Standardized all numeric-related fields.

② Filled Missing Values

Imputed missing entries across key variables.

③ Standardized Date Fields

Unified date formats and extracted basic time features.

④ Standardized Categorical Labels

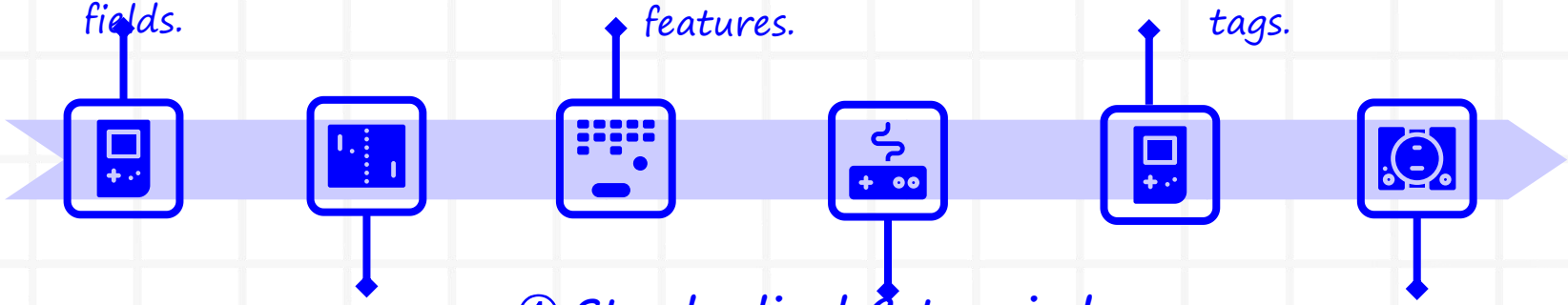
Cleaned and unified publisher/developer information.

⑤ Created Platform Indicators

Added PS3 / PS4 / PS5 identification tags.

⑥ Final Cleaning

Removed invalid values and exported the processed dataset.



Preparations: EDA & correlation & preliminary regression



Research Methods

The Gene Decoder: Four Analytical "Laboratories"

Signal Gene Analysis - The Magic of the "Trailer

Theory

- In the highly information-asymmetric video game market, signaling theory argues that publishers and developers use observable cues—such as brand reputation, advertising, media ratings, and market attention—to convey their otherwise unobservable product quality to players.
- Building on this theoretical foundation, we employ **OLS regression, interaction models, and random forest analyses** to empirically test which signals most strongly shape player expectations and ultimately drive sales.



Signal Gene Analysis - The Magic of the "Trailer Effect"

Key Variable Definitions

Dependent Variable, Y	log_sales	Log total sales, used to reduce right-skewed distribution.
Independent Variables, X	metacritic_signal	Metacritic score of the game (missing values imputed as 70), representing external expert assessment.
Independent Variables, X	quality_signal_strength	Normalized metric of perceived quality scaled to [-1, 1]: (metacritic−70)/30, enabling comparability across variables.
Independent Variables, X	publisher_reputation	Weighted index (sales 0.7, rating 0.3) capturing the publisher's historical performance, market influence, and trustworthiness.
Independent Variables, X	developer_reputation	Reputation index of developers using the same weighting method, reflecting creative capability and fanbase strength.
Independent Variables, X	user_anticipation	Log of player rating counts, serving as a proxy for early market attention, hype, and community anticipation.
Interaction Terms, X	quality_x_publisher	Captures whether publisher brand strength weakens or amplifies the impact of quality signals.
Interaction Terms, X	quality_x_developer	Tests whether developer reputation strengthens players' responsiveness to quality ratings.

Signal Gene Analysis - The Magic of the "Trailer

Model 1: Baseline OLS regression model

$$\log_sales = \beta_0 + \beta_1 * metacritic_signal + \beta_2 * publisher_reputation + \beta_3 * developer_reputation + \varepsilon$$

Model 2: Interaction-Enhanced Model

$$\log_sales = \beta_0 + \beta_1 * metacritic_signal + \beta_2 * publisher_reputation + \beta_3 * developer_reputation + \beta_4 * quality_x_publisher + \beta_5 * quality_x_developer + \varepsilon$$

Model Fit Comparison

Model	R^2	Adjusted R^2	Key Insight
Model 1	0.3441	0.3433	Baseline explanatory power
Model 2	0.3550	**0.3536**	Interaction terms improve model fit

Adding interaction terms increases the adjusted R^2 from **0.3433** → **0.3536**, indicating that **the relationship between quality signals and sales depends on publisher/developer reputation**. The model becomes more explanatory when incorporating brand–quality interactions.

Signal Gene Analysis - The Magic of the "Trailer

Regression Coefficient Interpretation

Variable	Coefficient	Significance	Interpretation
metacritic_signal	+0.0420	*** (p = 0.000)	Higher media scores increase log-sales; each 1-point increase ≈ +4.3% sales growth.
publisher_reputation	**+7.5686**	*** (p = 0.000)	For every 1 increase in publisher reputation, log-sales rise by **25.2%** (7.57/30), corresponding to roughly a 1.29-fold increase in actual sales. This indicates that **publisher reputation is one of the strongest drivers of market performance**.
developer_reputation	**+9.0353**	*** (p = 0.000)	For every 1 increase in developer reputation, log-sales rise by **30.1%**(9.04/30), corresponding to roughly a 1.35-fold increase in actual sales. **The influence of developer reputation is even stronger than that of publishers.**
quality × publisher_reputation	**-6.7491**	*** (p = 0.000)	There is a significant negative interaction between quality signals and publisher reputation: when publisher reputation is high, the marginal effect of review scores on sales becomes weaker. In other words, **strong publishers can rely on brand power even with average scores, while weaker publishers depend more heavily on high scores to gain market acceptance.**
quality × developer_reputation	+2.2073	n.s. (p = 0.091)	Developer reputation does not significantly alter the effect of media scores on sales.

Signal Gene Analysis – The Magic of the "Trailer Effect"

Random Forest:

The results of the random forest model are highly consistent with those of the linear regression model.

Feature	Importance	Key Insight
developer_reputation	**0.5938**	Strongest predictor of sales; **developer brand dominates market response.**
publisher_reputation	**0.2117**	Publisher brand significantly boosts sales; strong trust effect.
user_anticipation	**0.1413**	Early hype and community attention meaningfully predict demand.
metacritic_signal	**0.0532**	Weakest predictor; **review scores alone have limited power without brand context.**

Insight: The non-parametric nature of the random forest validates the robustness of the linear model. It reinforces that, **when predicting game sales, developer and publisher reputation are far more influential signals than the game's review score**.

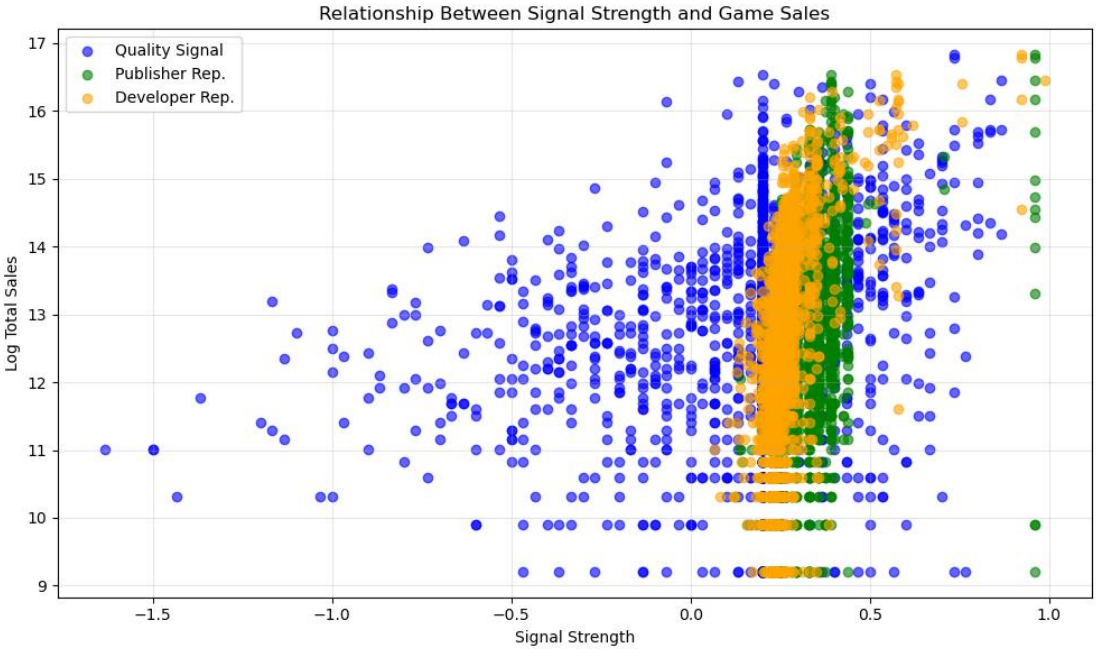
Signal Gene Analysis - The Magic of the "Trailer Effect"

Visualized Analysis:

Quality Signal: The distribution is wide and dispersed across the x-axis, showing no clear linear relationship with sales. This confirms that quality alone has weak predictive power.

Publisher Reputation: High-reputation games cluster in the upper sales region, indicating a strong positive association between publisher brand strength and commercial performance.

Developer Reputation: Developer reputation shows the clearest upward pattern; high-reputation developers consistently correspond to top-selling titles, making it **the strongest signal** among the three.



Signal Gene Analysis - The Magic of the "Trailer

Key Insights:

- **Brand reputation outweighs review scores**: Developer and publisher reputation are far stronger drivers of sales than media ratings. Well-known creators and publishers can secure strong sales even with average scores.
- **The marginal effect of quality decreases under strong brands**: High-reputation publishers can rely on brand trust, making players less sensitive to review scores. In contrast, low-reputation studios must depend on high scores to gain market acceptance.
- **Developer reputation holds dominant influence**: Across both regression and random forest models, developer reputation consistently shows the highest predictive power, indicating players care more about "who made the game" than "who published it."
- **User anticipation is an effective early signal**: Rating counts and community engagement serve as strong leading indicators of demand, but inflated expectations can backfire if game quality fails to meet the hype.

Synergy Genes Analysis - The "Chemical Reaction" of Golden

Brand alliance theory: *by quantifying the "synergy effect" generated through the collaboration between publishers and developers, it reveals which brand combinations can deliver market performance far beyond expectations.*

Variable Definitions

Variable	Definition
Synergy Score	Measures the gap between the actual performance of a collaboration and the expected performance based on each party's independent results.
Actual	The average sales of all games produced by the specific publisher–developer pair.
Expected	The average of the publisher's independent performance and the developer's independent performance.
Superlinear Value	Actual − Expected, representing the absolute incremental gain created by the collaboration.
Collaborations	The number of games jointly produced by the publisher and developer.

Formula:

$$\text{Synergy Score} = (\text{Actual} - \text{Expected}) / \text{Expected}$$

Synergy Genes Analysis - The "Chemical Reaction" of Golden

Top 5 Synergistic Pairs

Rank	Publisher + Developer	Synergy Score	Superlinear Value	Actual Sales	Expected Sales	Collaborations
1	Sony Interactive Entertainment + Insomniac Games	17.567	5,270,000	5,570,000	300,000	2
2	Namco Bandai Games + Arc System Works	12.905	1,355,000	1,460,000	105,000	1
3	Sony Computer Entertainment + From Software	10.626	2,842,500	3,110,000	267,500	1
4	Electronic Arts + EA Sports	7.470	4,052,500	4,595,000	542,500	2
5	Namco Bandai Games + From Software	6.583	987,500	1,137,500	150,000	4

- Sony Interactive Entertainment + Insomniac Games show the strongest synergy, with a synergy score of 17.567, meaning **their actual sales are 18.567 times the expected sales**.
- Namco Bandai Games + Arc System Works and Sony Computer Entertainment + FromSoftware also demonstrate very strong synergy, with synergy scores of **12.905** and **10.626** respectively.
- Although Electronic Arts + EA Sports have a lower synergy score (**7.470**), their superlinear value reaches **4,052,500**, indicating a substantial absolute increase in sales generated by their collaboration.
- Namco Bandai Games + FromSoftware have the highest number of collaborations (4), but their synergy score is relatively lower (**6.583**), suggesting that their partnership produces stable performance without explosive growth.

Synergy Genes Analysis – The "Chemical Reaction" of Golden

Synergy Effect Statistics

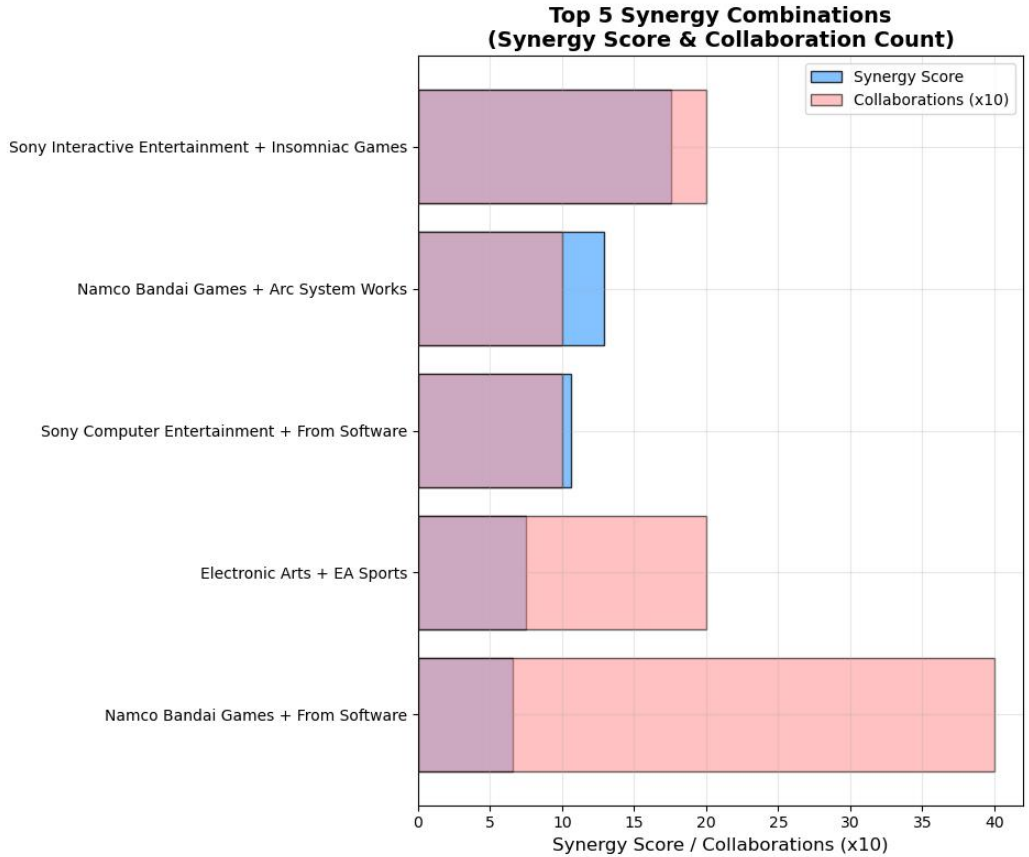
Metric	Value
Total Collaboration Pairs	1057
Positive Synergy Pairs	491 (46.5%)
Average Synergy Score	0.163
Maximum Synergy Score	17.567

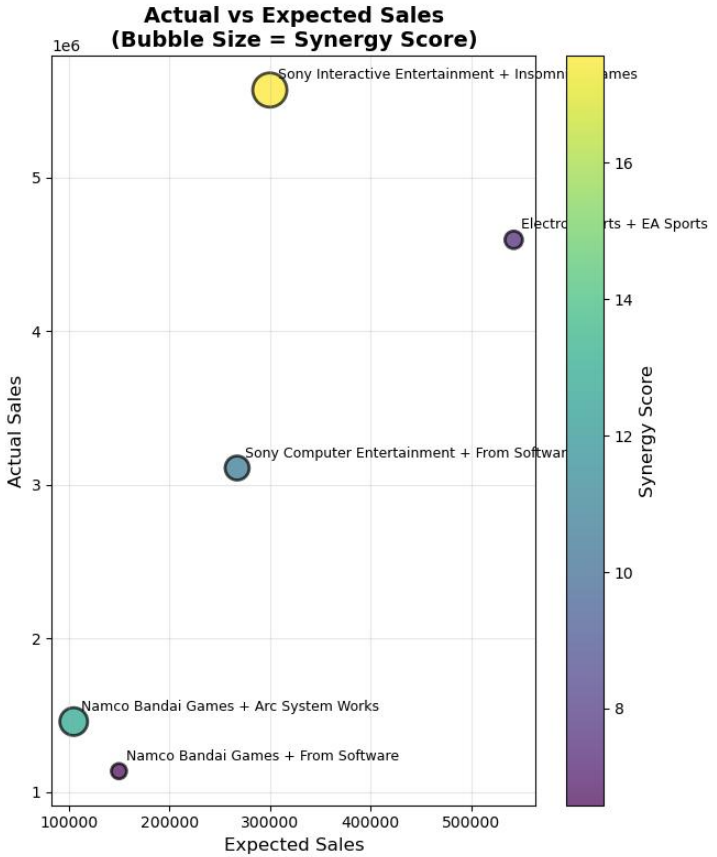
Insight:

- **Nearly half of all collaboration pairs (46.5%) generate positive synergy**, indicating that brand alliances are effective in the gaming industry and often create value beyond what each party can achieve independently.
- The average synergy score is 0.163, meaning that **on average, publisher–developer collaborations perform about 16.3% better than expected**, reflecting a consistent positive cooperation effect.
- The maximum synergy score reaches 17.567, showing that certain brand combinations can deliver market outcomes far exceeding expectations, producing exceptionally strong synergistic effects.

Synergy Genes Analysis - The "Chemical Reaction" of Golden

Visualization





Synergy Genes Analysis - The "Chemical Reaction" of Golden

Key Insights:

- **Synergy is a key driver of success in the gaming industry**: Publisher–developer pairs with high synergy scores consistently outperform expectations, demonstrating that shared resources and complementary strengths produce outcomes far greater than what either party could achieve independently.
- **Top synergistic partnerships rely on strong branding and innovation**: Top synergistic pairs typically combine strong brand equity with high innovation capability, allowing their collaborations to achieve exceptional market impact. This highlights how brand strength and innovation jointly amplify player attention and drive superior sales performance.
- **Synergy exhibits a scale effect—the more collaborations, the greater the efficiency**: As partnerships deepen, improvements in coordination, workflow, and resource integration lead to stronger synergy and increasingly efficient market performance.
- **Synergy shows regional variation, performing differently across markets**: Because player preferences and cultural factors differ by region, the effectiveness of publisher–developer partnerships is not uniform, underscoring the need for localized strategies tailored to specific markets.

Diffusion Genes Analysis - Survival Wisdom in "Red & Blue

Bass Model: provides a classic quantitative tool, categorizing adopters into two main groups influenced by external factors (innovators) and internal communication (imitators), thereby predicting a product's diffusion trajectory and market potential.

Variable Definitions

Variable	Definition
Cumulative Adoption, F(t)	Cumulative sales at time t, used to fit the diffusion curve.
Time Period, t	Time since game release.
p – Innovation Coefficient	Innovation coefficient measuring early adopter influence. (Estimated)
q – Imitation Coefficient	Imitation coefficient measuring social contagion. (Estimated)
m – Market Potential	Maximum possible cumulative sales in a region. (Estimated)
p + q – Adoption Speed	Adoption speed; higher values indicate faster diffusion.
peak_time	Time of peak adoption.

Formula:

$$F(t) = m * [1 - \exp(-(p+q)t] / [1 + (q/p) * \exp(-(p+q)t]$$

Diffusion Genes Analysis - Survival Wisdom in "Red &

Regional Diffusion Parameters

Region	Innovation Coefficient	Imitation Coefficient	Adoption Speed	Market Potential
NA	0.0010	0.0010	0.0020	584,740,000
PAL（Europe）	0.0010	0.0010	0.0020	580,530,000
JP	0.0010	0.0017	0.0027	107,960,000
Other	0.0010	0.0010	0.0020	219,570,000

- **Japan (JP) shows the fastest adoption speed** (0.0027), significantly higher than other regions (0.0020). This indicates that **Japanese players adopt new games more quickly, leading to a faster diffusion curve.**
- **Japan's imitation coefficient (q = 0.0017) is notably higher** than that of all other regions (q = 0.0010), suggesting that **Japanese consumers are more likely to purchase games based on word-of-mouth or social influence.**
- **North America (NA) and Europe (PAL) have the largest market potential** (around 580 million), making them the primary battlegrounds for global game sales.
- **Japan has the smallest market potential (≈108 million)**, but its **sales efficiency per unit time is the highest**, driven by strong social contagion and fast adoption dynamics.

Diffusion Genes Analysis - Survival Wisdom in "Red &

Hofstede's Cultural Dimensions Theory states that countries differ systematically in values, which shape how quickly consumers accept new products. The theory argues that **greater cultural distance increases market-entry barriers and slows adoption,** making cultural distance a key factor in explaining regional adoption lag.

Region Pair	NA-JP	NA-PAL	NA-Other	JP-PAL	JP-Other	PAL-Other
Cultural Distance	99.73	28.14	61.20	91.64	65.30	62.08

- NA–JP has the largest cultural distance (99.73), indicating substantial cultural differences between the two markets.
- NA–PAL has the smallest cultural distance (28.14), suggesting that North America and Europe are culturally similar.
- JP–PAL shows a large cultural distance (91.64), indicating significant cultural differences between Japan and Europe.

Insight: Because of the large cultural gap, Japanese games face greater difficulty entering Western markets and require stronger localization and market research. In contrast, European games can more easily enter the North American market due to cultural similarity.

Diffusion Genes Analysis - Survival Wisdom in "Red &

K-means Clustering Analysis

Cluster analysis based solely on each region's sales share:

Cluster ID	Number of Games	Dominant Region	Example Titles
Cluster 0	918	NA	Ridge Racer 7, Call of Duty 3, Need for Speed Carbon
Cluster 1	463	JP	Sega Golf Club, Mobile Suit Gundam: Crossfire, Mahjong Taikai IV
Cluster 2	593	NA	NHL 2K7, NBA 2K7, Tiger Woods PGA Tour 07
Cluster 3	261	PAL	Formula 1: Championship Edition, DiRT, FIFA Soccer 08

- **North America dominates the largest number of games** (Cluster 0 and Cluster 2, totaling 1,511 titles), followed by Japan (Cluster 1, 463 titles), and finally Europe (Cluster 3, 261 titles).
- **Japanese-dominated game clusters feature distinct genres**, mainly sports, racing, and mahjong-style games, reflecting the strong alignment with Japanese player preferences.
- **North American–dominated clusters include a wide variety of genres**—racing, shooters, sports, action-adventure, and more—indicating that North American players have broader and more diverse tastes.

Diffusion Genes Analysis - Survival Wisdom in "Red &

K-means Clustering Analysis

Incorporating game genre as a key variable into the clustering analysis:

Cluster ID	Number of Games	Dominant Region	Dominant Genres	Example Titles	Example Genres
Cluster 0	918	NA	Shooter	Call of Duty 3, F.E.A.R., The Darkness	Shooter, Adventure, Action
Cluster 1	463	JP	Action	Sega Golf Club, Mahjong Taikai IV, Railfan	Unknown, Strategy
Cluster 2	593	NA	Simulation	Ridge Racer 7, Need for Speed Carbon	Racing, Arcade, Simulation
Cluster 3	261	NA	Action	NHL 2K7, NBA 2K7, Tiger Woods PGA Tour 07	Unknown, Action, RPG

Insight:

- The North American market is "genre-diverse," covering multiple submarkets such as shooters, simulation, and sports-action. In contrast, the Japanese market is relatively "genre-concentrated," dominated by the broader action category, with characteristics that are lighter, more casual, and culturally distinctive.
- **For the North American market, a "multi-genre portfolio" strategy is required; for the Japanese market, a "deep localization" strategy is essential.**

Competition Gene Analysis - Survival Wisdom in "Red & Blue

Graph-based Market Competition Model: By constructing a game competition graph and applying graph neural networks together with density-based clustering, the model identifies market competition structures and blue-ocean opportunities.

Variable Definitions

- **Node**: Represents a game, including attributes such as game title, sales, media rating, and more.
- **Edge**: Represents the competitive relationship between 2 games; the edge weight reflects the substitution probability (i.e., the likelihood that players choose between the two games).
- **Substitution Probability**: Calculated based on feature similarity and a time-decay factor.
  - **Feature Similarity (cosine similarity)**: Measures how similar two games are in terms of media score, user score, publisher reputation, developer reputation, game genre, etc.
  - **Temporal Decay**: Captures how close the release dates of the two games are—the closer the release time, the higher the substitution probability. The formula is exp(-time_diff/12), where time_diff is the difference in release dates between the two games measured in months.

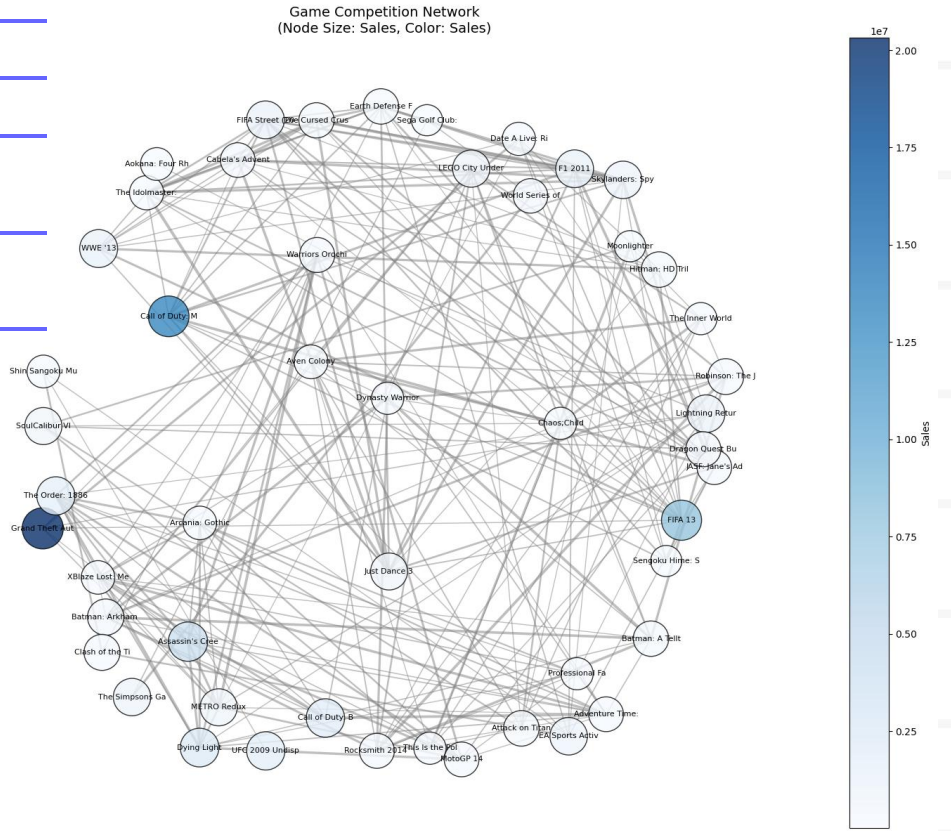
Formula:

$$\text{Substitution Probability} = \text{Feature Similarity} \times \text{Temporal Decay}$$

Competition Gene Analysis – Survival Wisdom in "Red & Blue Ocean"

Metric	Value
Total Number of Games	2240
Sampled Games	300
Number of Nodes in the Competition Graph	300
Number of Edges in the Competition Graph	8426

- Node size and color represent game sales: games with higher sales appear as larger and darker nodes, such as Grand Theft Auto V and Call of Duty: Modern Warfare.
- **The competition graph is highly dense**, with each node having an average of 28 edges (8426/300), indicating that the game market is highly competitive and players choose among many alternatives.

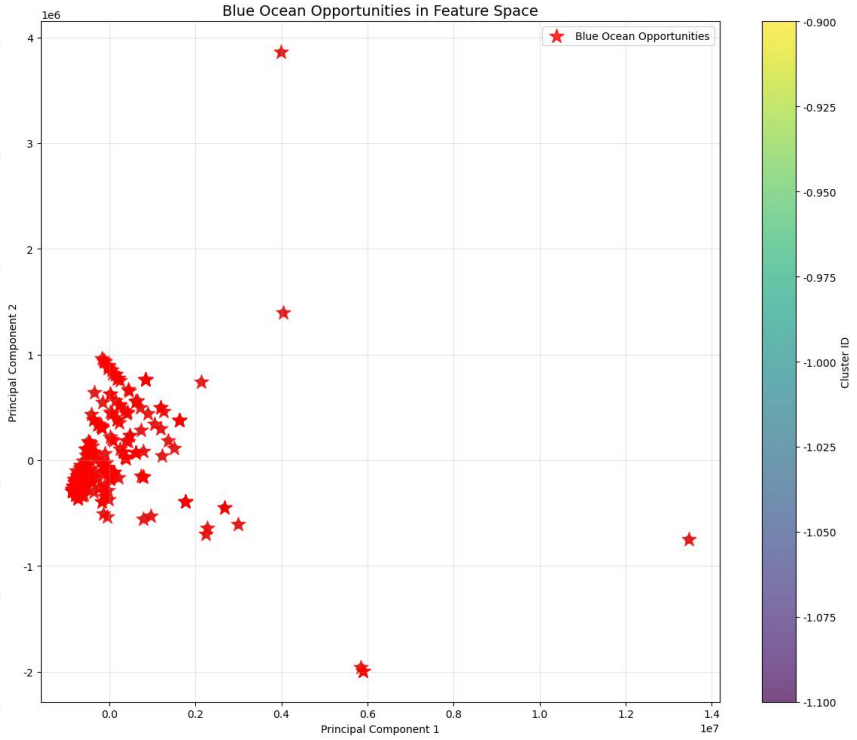


Competition Gene Analysis - Survival Wisdom in "Red & Blue

Identifing "Blue-ocean Opportunities": Using DBSCAN density clustering: games classified as noise points—those not belonging to any cluster—are treated as isolated products in the feature space, indicating underexplored or underserved market segments.

Characteristics of blue-ocean opportunities:

- **Lower sales** (e.g., This Is the Police 2 with only 20,000 units sold).
- **Moderate media ratings** (e.g., The Simpsons Game with a score of 70).



Gene Decoder Integration: Activating the "Ultimate Synthesizer"

Multiple Linear Regression: Quantifying the Key Drivers of Game Sales

Key Variable Definitions

Variable	Definition
log_sales	Log of total game sales, used to correct right-skewed distribution.
metacritic	Media critic score of the game.
rating	User rating score.
ratings_count	Number of user ratings (user anticipation signal).
publisher_reputation	Reputation score of the publisher.
developer_reputation	Reputation score of the developer.
release_year	Year of game release.
release_month	Month of game release.
genre_X	One-hot encoded game genres (e.g., *genre_Action*, *genre_Adventure*, etc.).

Formula:

$$\text{log_sales} = \beta_0 + \beta_1 * \text{metacritic} + \beta_2 * \text{rating} + \beta_3 * \text{ratings_count} + \beta_4 * \text{publisher_reputation} + \beta_5 * \text{developer_reputation} + \beta_6 * \text{release_year} + \beta_7 * \text{release_month} + \Sigma\beta_i * \text{genre_i} + \varepsilon$$

Overall Model Fit: Adjusted $R^2 = 0.470$

Gene Decoder Integration: Activating the "Ultimate

Regression & VIF Results

Variable	Coefficient	Significance	VIF	Interpretation
const	174.3367	***	0	Baseline log-sales when all predictors are zero.
rating	0.2355	***	1.270052	Each +1 user rating → sales ↑ **26.5%**. Strongest sales driver.
ratings_count	0.0004	***	1.648685	Each additional rating → sales ↑ **0.04%**. User anticipation matters.
publisher_reputation	4.122e-07	***	1.688561	Higher publisher reputation slightly increases sales.
developer_reputation	4.173e-07	***	1.768694	Higher developer reputation slightly increases sales.
release_year	−0.0859	***	1.195931	Newer games sell **8.2% less** per year (older titles accumulate sales).
genre_X	8.5254 ~ 10.4625	***	INF	The one-hot encoded genre variables create perfect collinearity, resulting in infinite VIF.

Gene Decoder Integration: Activating the "Ultimate

Key Insights:

- **User ratings are the strongest driver of game sales:** A one-point increase in user rating leads to roughly a 26.5% increase in sales, highlighting the central role of perceived quality. This suggests that improving game quality and enhancing player satisfaction should be a **primary strategy** for developers seeking to boost commercial performance.
- **Game genre has a substantial impact on sales:** with categories such as card, fighting, sports, and family games demonstrating the highest performance. These results indicate that genre selection is a critical strategic decision, and developers may benefit from focusing on **high-demand genres to maximize sales potential**.
- **Publisher and developer reputation both have positive effects on sales:** although the magnitude is smaller compared to ratings and genre. Even so, reputable partners enhance credibility and reduce perceived risk for players. Developers can therefore improve expected sales outcomes by collaborating **with well-established publishers and studios**.
- **Later release years are associated with lower cumulative sales** with each additional year reducing sales by approximately 8.2%. This reflects the advantage held by earlier titles in accumulating long-term sales. To stay competitive, developers should closely monitor market trends and release products **in a timely manner** to

05

Summary

Conclusion, Innovation and Managerial Implications

Conclusion

Ushering in the "Genetic Evolution New Era"

Research Framework

Theoretical Integration: Signal Theory、Brand Alliance Theory、Innovation Diffusion Theory、Graph Neural Networks (GNNs)

Multi-Dimensional Market Response Model: A dynamic, data-driven framework revealing the "invisible market logic" of the gaming industry.

Key Findings

- User Ratings → the Strongest Driver of Sales
- Brand Synergy Effects → Generate over 18 times the Excess Returns
- Game Genres + Regional Diffusion Patterns → Jointly Shape Competitive Landscape

Methodological Innovation

Building the "Gene Decoding Laboratory"



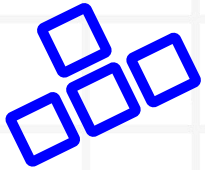
"Game Genetic Sequencing Platform": Constructing a multi-level decoding framework.

This framework treats game market success as the result of the combined action of four genetic dimensions:

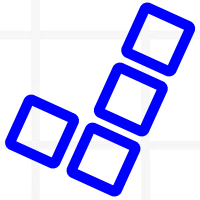
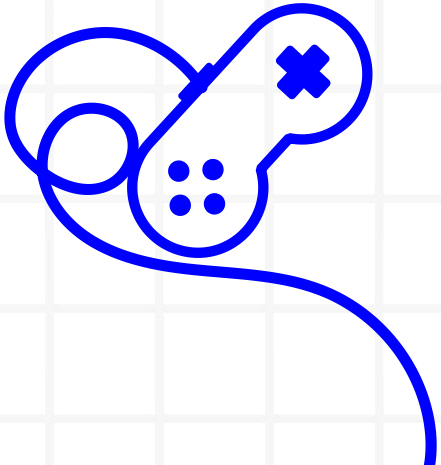
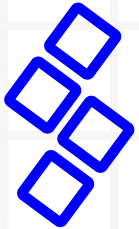
Market performance = f(signal gene × diffusion gene × Synergistic gene ×

Theoretical Contributions & Practical Implicat

- Theoretical Contributions
  - Integrated **signal theory**, **brand alliance theory**, and **innovation diffusion theory** to build a **dynamic market response model**.
  - First to **quantify brand synergy**, distinguishing **internal vs. external** synergy.
  - Applied **graph neural networks** to identify **blue ocean opportunities** in game competition.
- Practical Implications
  - **For Developers**
    - Prioritize **high-selling game genres**
    - Partner with **reputable publishers**.
    - Focus on **user ratings**.
  - **For Publishers**
    - Invest in the **Japanese market**.
    - Work with **high-quality developers** to boost synergy.
  - **Further research**
    - Use **GNNs** to find blue ocean segments.




THANKS

MKT6019 Group 5